Project Documentation: Cloud Drive

Jesse Mäkelä
jesse.j.makela@student.lut.fi

1. Introduction

Cloud Drive is a web-based document management system designed to function similarly to services like Google Drive or OneDrive. It allows users to create, edit, rename, and delete text documents while also supporting the uploading of image files. Key features include sharing files with specific users through edit or view permissions, generating public links for non-registered users, and a locking mechanism that ensures only one person can edit a document at a time. Users can also personalize their experience by setting profile picture.

2. Technology choices

The project is built using these technologies

- **Backend:** Node.js.
- **Database:** MongoDB for persistent data storage.
- **Frontend:** TypeScript

3. File structure

This is screenshot of text file made by GitHub copilot:

```
Project/
├── backend/          # Node.js + Express API (port 3001)
│   ├── server.ts
│   ├── uploads/
│   │   ├── images/   # Uploaded drive images
│   │   └── avatars/  # User profile pictures
│   └── src/
│       ├── models/       # Mongoose schemas (User, Document, DriveImage)
│       ├── routes/       # API route handlers (auth, documents, users, files)
│       ├── middleware/   # JWT auth middleware
│       └── validators/   # Input validation rules
└── frontend/         # React frontend (port 5173)
    └── src/
        ├── pages/        # Login, Register, Dashboard, Editor, PublicDocument
        ├── context/      # AuthContext (global auth state + refreshUser)
        ├── components/   # ProtectedRoute
        └── api/          # Axios client (authAPI, documentAPI, fileAPI, userAPI)
```

4. Implemented features and points

Feature	Description	Points
Basic Features	Authentication, CRUD operations, sharing, edit-lock, and responsive design.	25
Frontend Framework	Utilization of a modern frontend framework (React/TypeScript).	3
Profile Picture	Users can select and store a profile picture on the server.	2
Image Upload	Users can upload image files to their drive.	2
Total Aimed Points		32

5. AI decleration

AI Systems Used: [NoteBookLM, GitHub Copilot].

Usage Description:

- **Debugging:** Used GitHub Copilot to identify and fix logic errors in backend routes
- **Code Generation:** For example, used to generate boilerplate for Express routes.
- **Proofreading:** For example, used to check the grammar of this documentation.